WiFi over larger areas. Technically speaking, WiFi will often be used for local (home) networks, whereas WiMAX will be used for last mile connectivity (between your home and ISP)—thankfully no more dug up roads!

In this comparison, we have ten routers—five based on 802.11g and five on 802.11n, respectively. Those based on 802.11n are backward compatible with 802.11g, but not the other way round. Though 802.11n-based routers provide faster data speeds, they are plagued with compatibility issues and may not deliver the performance they should when used with laptops or devices using 802.11g wireless cards. From here on, 802.11g-based devices will be referred to as the 'g' group, while 802.11n-based ones will be called the 'n' group.

Features

Aesthetics

If you plan to install a WiFi router in your living room, you surely don't want something that is finished in grey plastic, looks chunky and has many antennas staring in your face.

In the 'g' group, Netgear's WGR615 has a pearl-white finish and looks appealing. This is followed by ASUS, which also employs a similar finish. D-link has come a long way from the grey plastic boxes they used to offer. The newer routers look better in the black-silver outfit. Linksys' WRT54G wouldn't qualify in our best looking router though. Compex NP25G still reminds us of the 33 Kbps Modem days.

In the 'n' group the Linksys WRT310N simply blew us away. It doesn't look like a router from any angle. We reckon this is how routers should be designed! Netgear's WNR834B rectangular looks didn't impress us much. The D-link trio were nearly identical, except for the top-end DIR-655, which was finished in white.

Connectivity

Routers come with loads of connectivity options, Ethernet ports, USB ports and wireless of course! Getting the right kind of device as per your need is also the key. In the 'g' group, the ASUS WL-520GU was the only router to come with USB port, and offers print server capability. D-link's 635 and 655 in the 'n' group were equipped with USB ports. These routers currently don't support a print server, but it is expected in D-link's new firmware due to be released in a few months. For now, you can just use them to exchange wireless settings using Windows Connect Now (WCN).

Depending on the radio configuration, the number of antennas varies from product to product. Some vendors offer routers with no external antennas. However, they do have them concealed to improve aesthetics. Linksys' WRT310N is one such device, and without the external antenna, it performed well in our tests. D-link's DIR-655 on the other hand, had three of them and returned good results. However, there wasn't significant difference between the two. All g-based routers had just one antenna and they performed well.

Device Features

D-link's DIR-655 has a blue backlight with icons on a silver panel, which affects legibility from a distance. In our opinion, wall mounting is an important feature, because mounting a router higher up ensures that signals are not attenuated (weakened) by furniture, doors, metal objects and so on. Netgear's WGR615

How We Tested

Test Setup

Each WiFi device was setup in an identical, elevated position and was kept away from metal objects, thick wooden walls to minimise interference.

The testing was done in two zones. Distances aren't important for benchmarking any sort of WiFi network, it's the obstructions between the access point (AP) and the signal recipient that ultimately plays the lead role. With this in mind, we designed each of these zones to actually take these wireless devices to their performance limits.

Zone 1: The laptop was placed within 20 feet of the AP. There was a wooden partition between AP and client through they were essentially in the same room. There was no shortage of signal attenuating obstructions.

Zone 2: We placed the laptop within 35 feet of the router. This time we ensured a decent load of signal-obstructing objects. There was a concrete wall (with cupboards) between both the WiFi devices to scramble the reception. There were also a couple of wooden partitions to help.

Test Process

File Transfer

We took two 400 MB files, one a single RAR file and the other a folder full of MP3s and video clips. The files were stored on a PC hard drive, which was connected to the Ethernet ports of the router. These files where then copied on a laptop at the two zones, and the time taken was noted down. Before beginning the transfer, we noted the signal reception strength in each zone.

Movie Streaming

Next, we streamed an HD video file encoded at 1280x720 (720p). This clip was present on our test PC's hard disk, and streamed to the laptop. In case any jerking of frames was really noticeable, points were deducted accordingly.

Test Laptop

We used Sony's VGN-FZ35GN notebook for this test–it comes with Intel's 4965AGN–compatible with both 2.4 and 5 GHz frequencies.

D-link's DIR-655

Linksys WRT310N

was the only router to skip this feature. All routers came with good bundle featuring, one Ethernet cable, power adapter, a quick install guide and one CD featuring the installation wizard.

Installation

For a lay person, configuring routers can be a daunting task—you to know at least some networking basics. However, these days most routers are bundled with wizards that make installing them a breeze.

Linksys, ASUS, Netgear, D-link and Compex, each have their version of wizards to help you out. You need to run the accompanying CD before connecting the router to the PC. After a few rounds of questions, the device is set. For geeks, these routers are a treasure trove—days of fun tinkering around with the settings. During our testing none of the routers needed any special attention while installing.

Web Server Interface

This provides the interface to micromanage the router's functionality. All settings are done via the Web server. A lot depends on how this interface is designed—grouping of key menus, granularity of settings, etc.

Linksys devices are really simple to use. D-link has done away with their old, quirky interface and has come up with a new, superb looking and easy-to-use interface. Netgear's interface has an excellent menu structure, but needs some polishing. One thing that sets Netgear apart is the ready availability of help. There is brief explanation of each setting and what they do. Great for someone lacking technical expertise. The Compex interface is bland and will probably confuse the newbie.

Security

Having the best possible protection is a must when it comes to wireless networks. Almost all the routers that we reviewed, support WPA2-PSK with AES or TKIP encryption. They also have basic integrated SPI (stateful packet inspection) firewalls to safe guard the network. Though all routers had these features, some such as Linksys offer predefined rule sets, which can be applied with a single click. In others, one has to work a little harder to make it happen.

Network Features

DHCP was up and running on all routers by default, and all of them give options to limit the number of client connections. Advanced features such as NAT and port forwarding for specific applications are also available. Blocking certain URLs, services and applications is also possible. In Linksys, D-link and Netgear, these settings are easy to work with, simply because of the intuitive interface. Each router allows only specific number of rules to be set—not more than 15 in all cases.

Device Management

Basic device management such as changing passwords, rebooting the router, running diagnostic tools and upgrading the firmware can be done on all routers we reviewed. Router set-

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